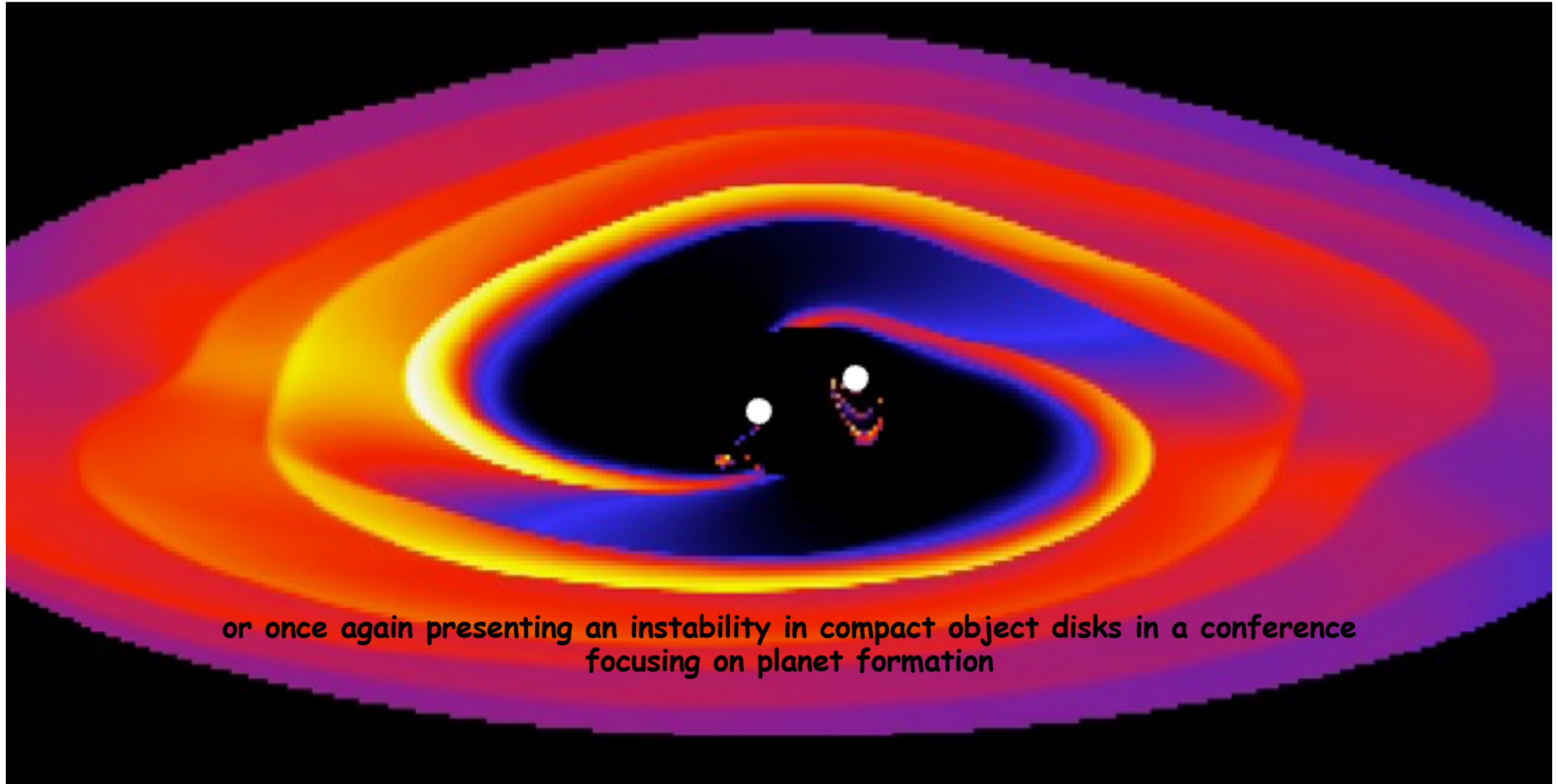




The Rossby Wave Instability at the edge of circumbinary disks



or once again presenting an instability in compact object disks in a conference focusing on planet formation

why do we look for instabilities?

they provide a reason for something that theory needs

- like the origin of angular momentum transport with the MRI

we are looking for an explanation of something we observed

- like periodicities in lightcurves of low mass Xray binaries

another key feature is that they offer a reason why something happens at some times and not others.

→ driven by a need in theory or explaining an observable

more recently a new need arises, that melds both aspects

→ need to find the best observable to detect pre merger BBHs

searching for binary black-hole

finding a pre merger BBH would allow us to follow the evolution of the system up to and after the merger

→ see how the strong and fast evolving gravity up to and after the merger affects the dynamical evolution of the accretion/ejection structures

problem: GW detection gives a quite large sky localization for the source

→ if gas is present in the pre-merger system it can help to localize the system faster through EM telescopes

→ but for that, we need to know what a pre-merger BBH would look like

analytical work is not enough

→ need for numerical simulations

→ and even further there is a need for observables

obtaining observables require a lot more than standard (M)HD simulations. depending on the observables, there is dust particle accumulation and scattering, reflection... taking into account GR effects...

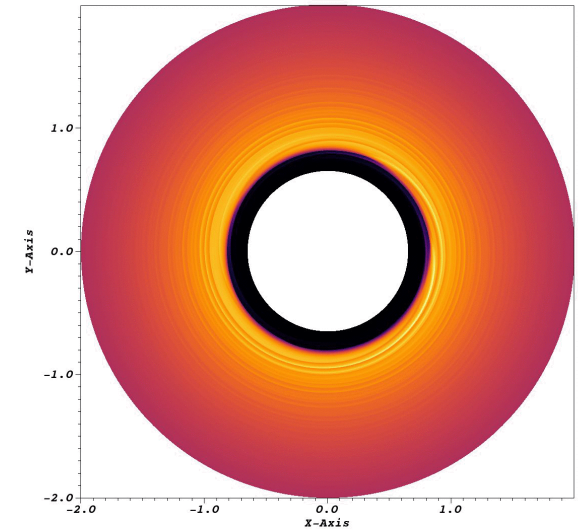
→ need for an extra step beyond fluid simulations

eNOVAs: extended Numerical Observatory of Violently Accreting systems

Fluid dynamical simulation performed with GRAMRVAC with a BBH metric

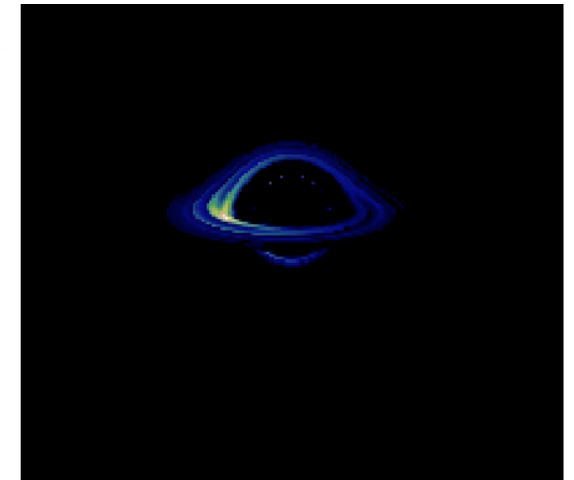
(Mignon-Risse, Varniere & Casse, 22):

- MPI parallelization
- block based AMR
- multi-dimensional simulations with cartesian or spherical grid
- GR (M)HD code with analytical metrics

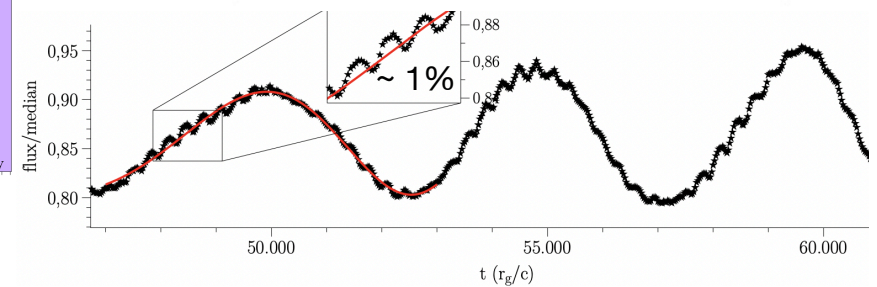
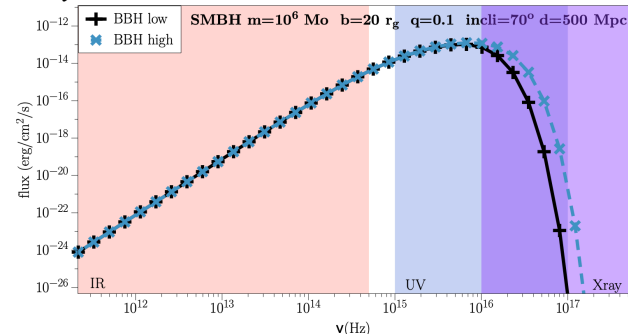


Ray tracing in full GR performed with gyoto

- backwards ray tracing from distant observer
- Runge-Kutta-Fehlberg adaptive-step integrator at order 7/8
- analytical metrics
- emission interpolated from GRAMRVAC



(using **SIXTE** we can produce instrument-dependent spectrum)

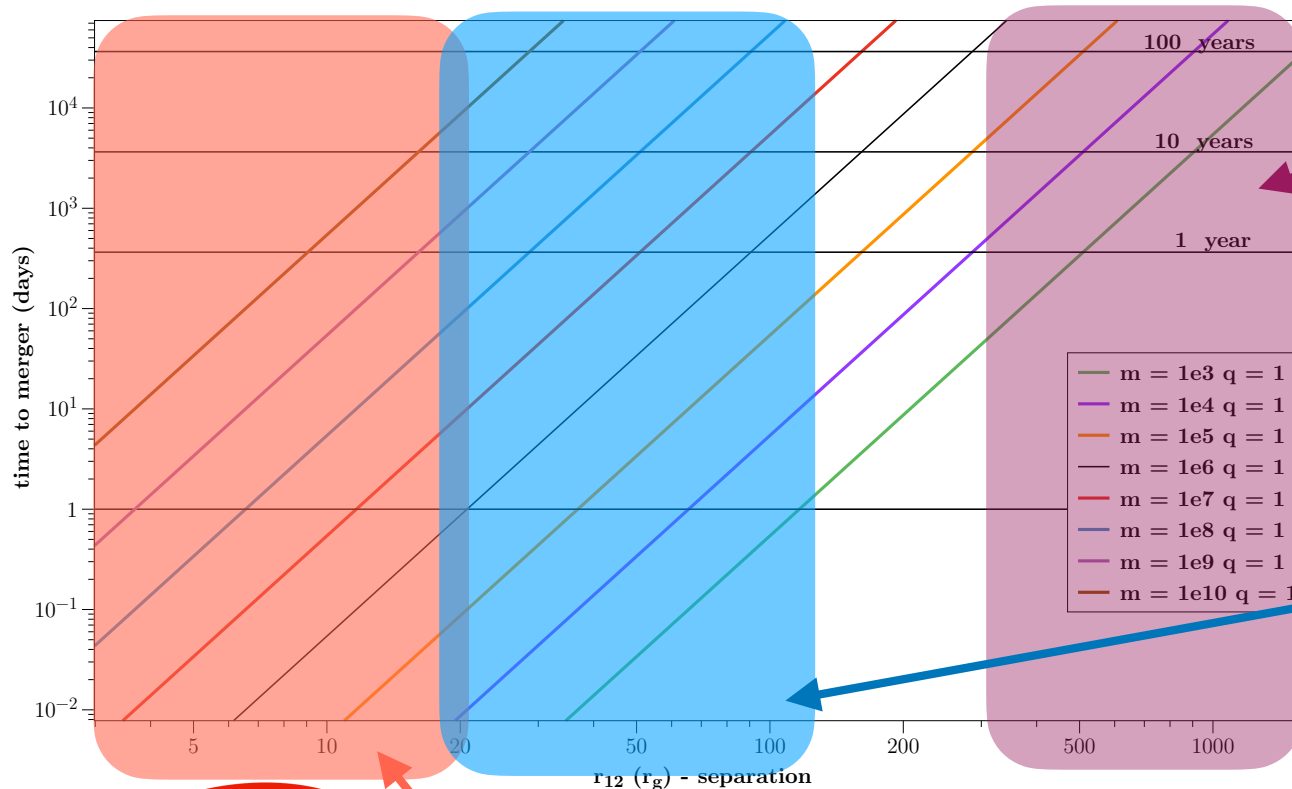
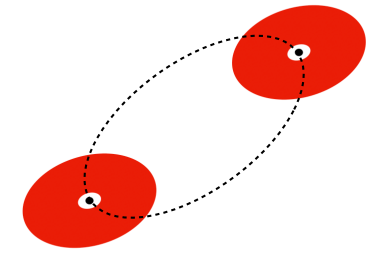


⇒ use your favorite tools to reduce the "observations"

the pre-merger BBH setup

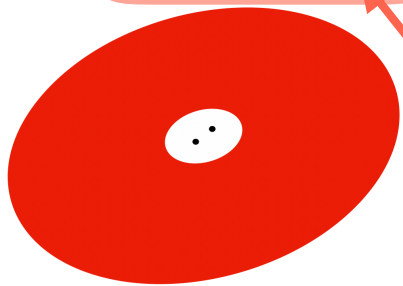
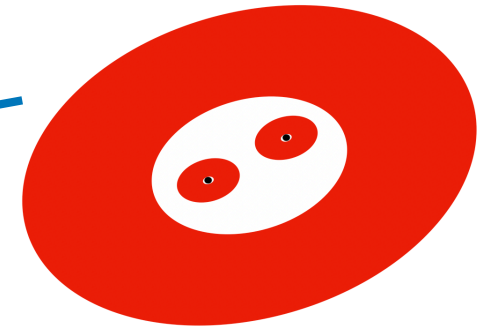
the **earliest stage** is two distant black-holes with their individual accretion disks

→ the slow inspiral impacts both disks and could help detect them



→ most of the systems at that stage **will not merge in the near future.**

as the black hole gets closer they gain a **circumbinary structure** on top of the two black-holes with their individual accretion disks



going even closer to the merger, the **inspiral prevents stable individual accretion disks**

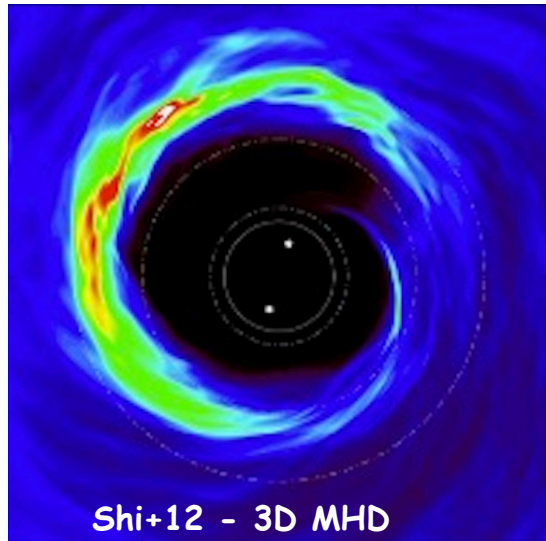
→ two close black-holes with a single, circumbinary, disk

→ since the individual accretion structures are **transient** as the black holes inspiral, the **circumbinary disk** is the best candidate to search for **any existing structures/observables** that could be related to the binary

the "lump" at the edge of equal-mass circumbinary disks

the problematic : equal mass binary,
a circumbinary disk,
two arms threading the cavity

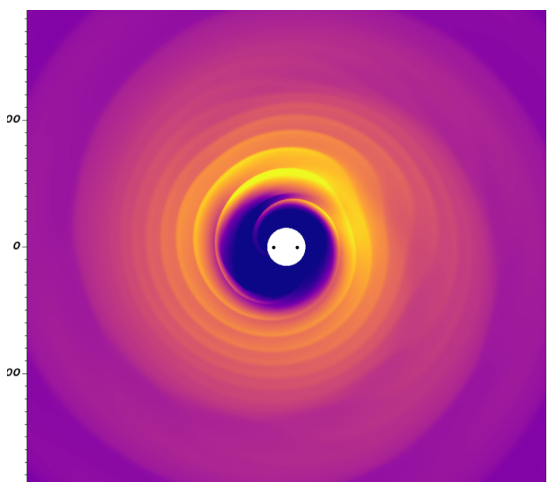
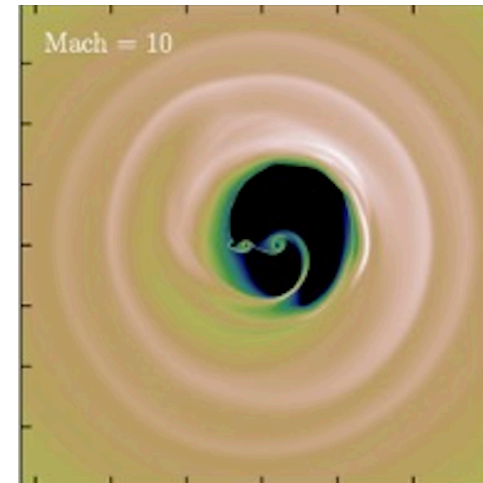
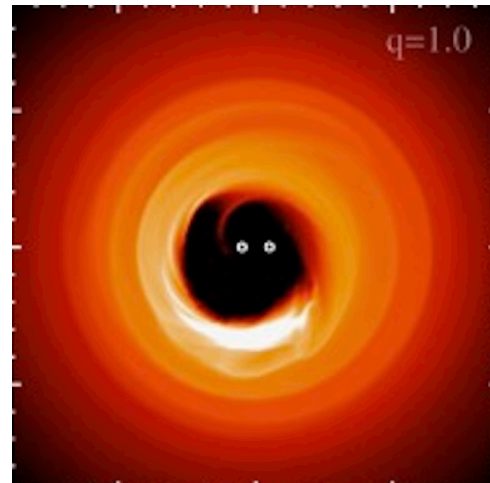
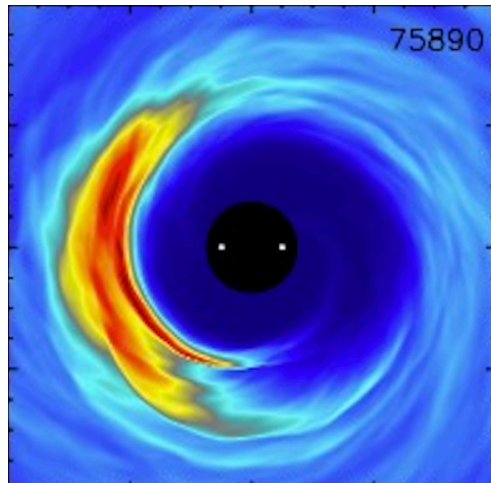
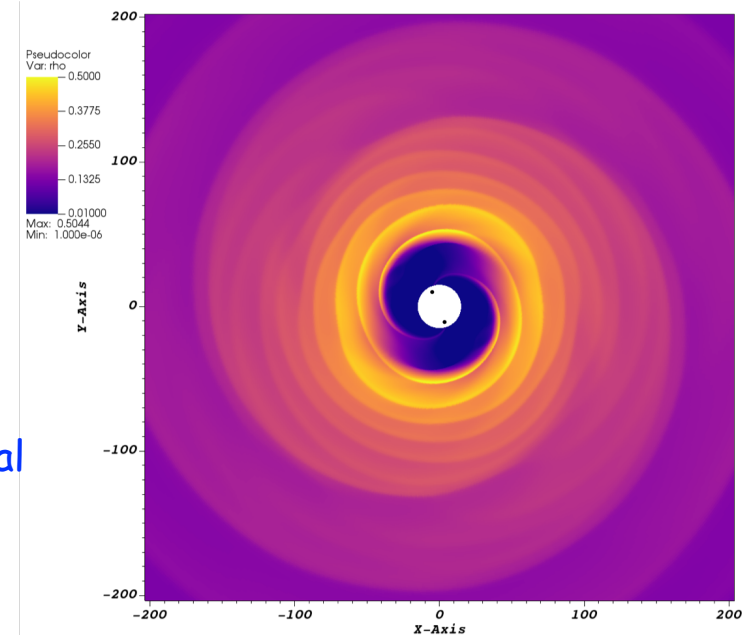
→ all is symmetrical



except it does not stay symmetrical

we see a "lump" being formed

→ this behavior has been reproduced for similar parameters by other groups and a few phenomenological causes were proposed for the lump



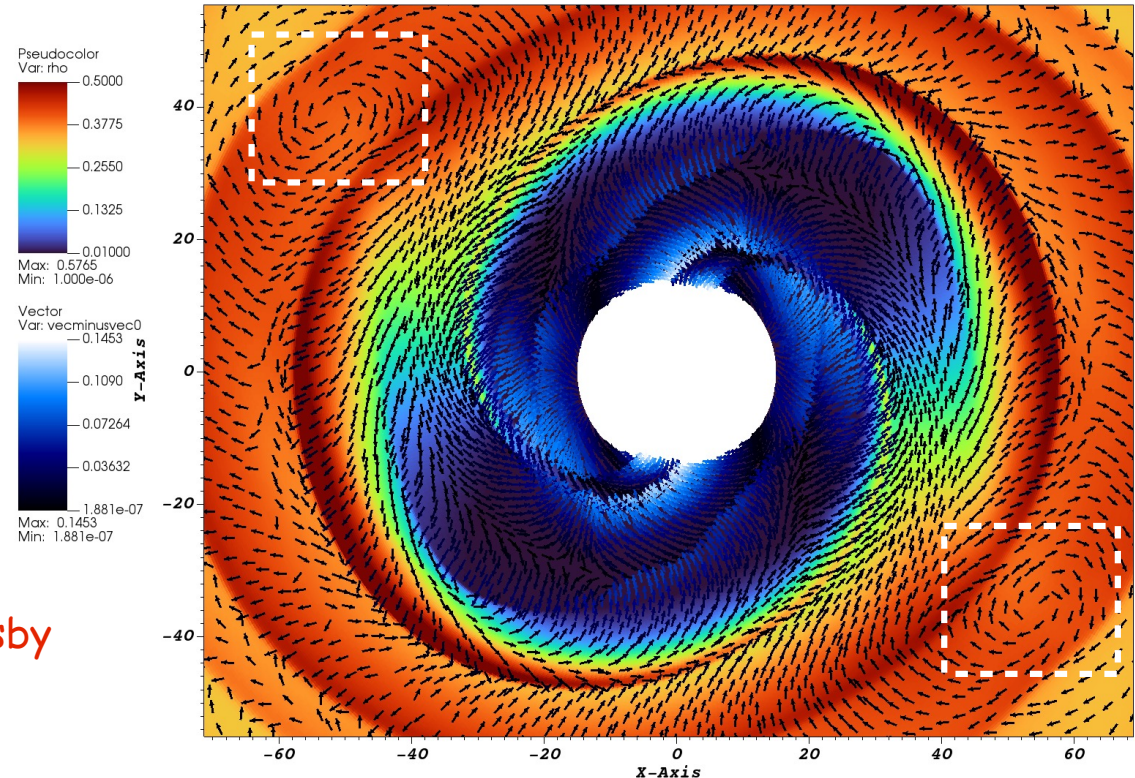
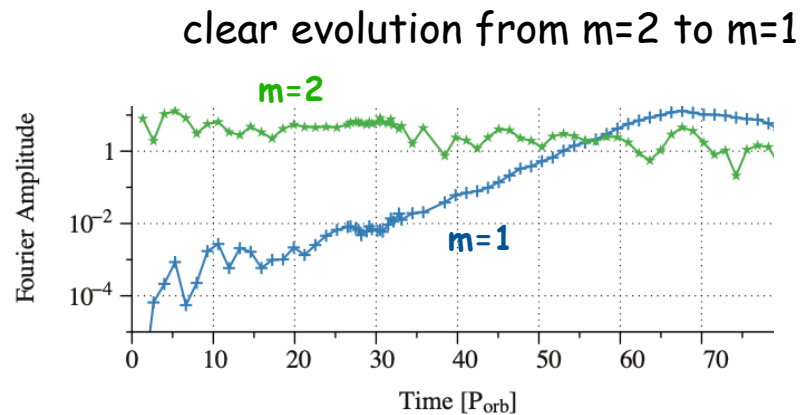
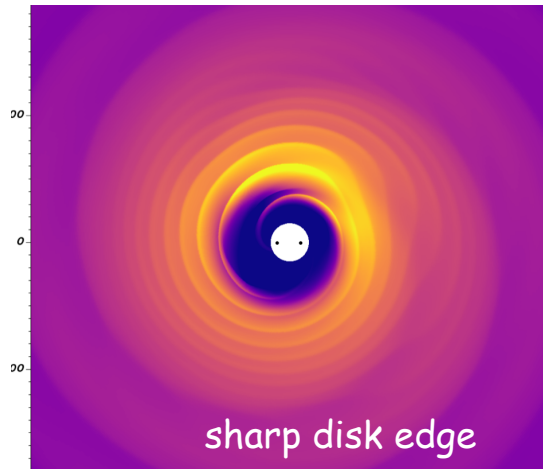
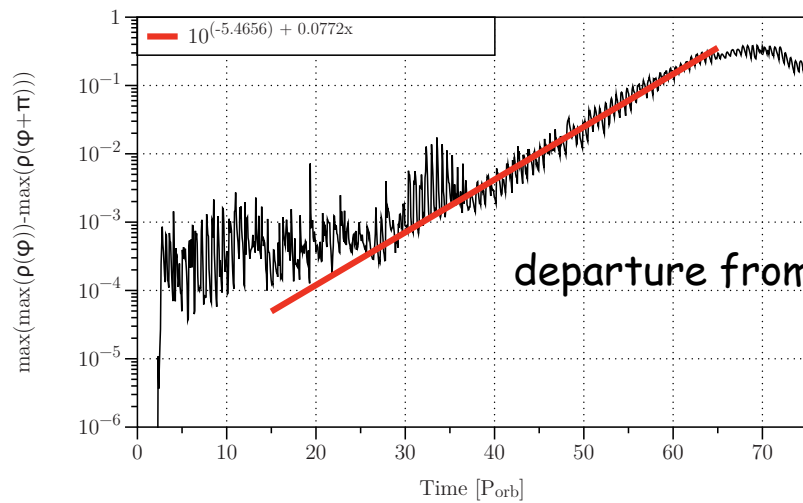
Noble+12 - 3D GRMHD

Ragusa+20 - 3D SPH

Tiede+20 - 2D HD

Mignon-Risse+23, 2D GRHD

the "lump" at the edge of equal-mass circumbinary disks



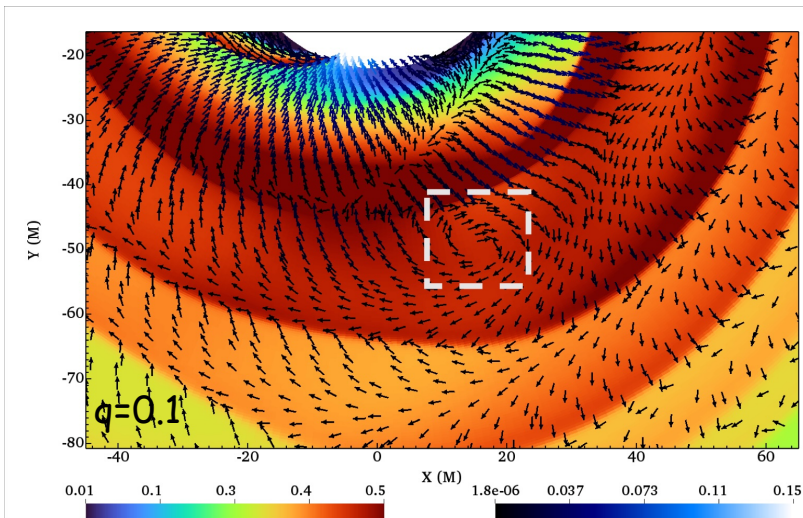
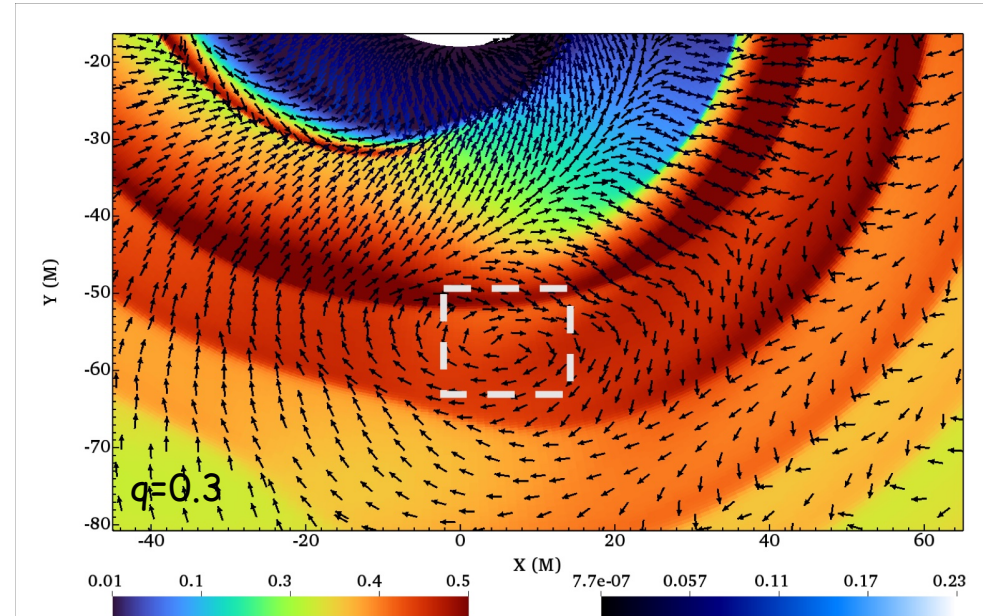
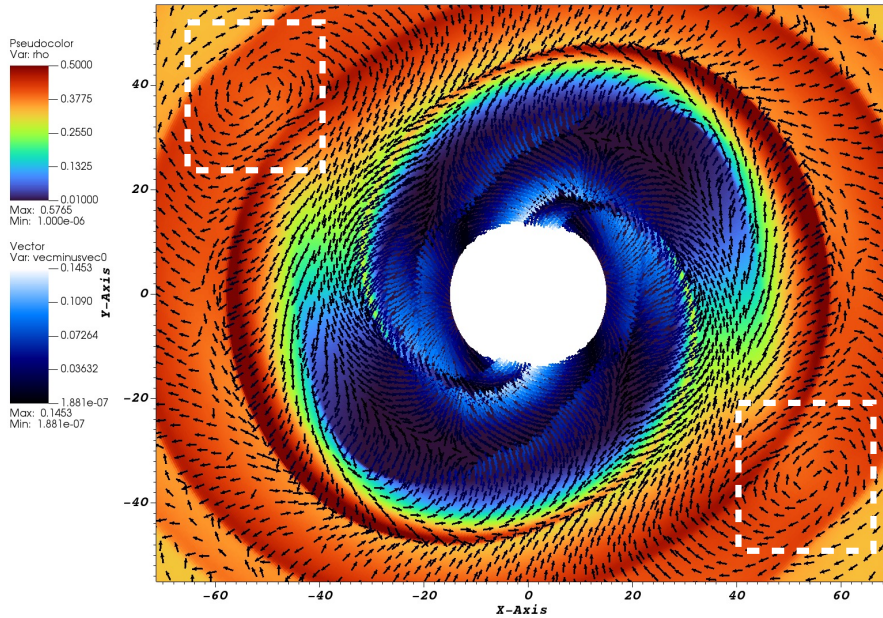
→ this behavior is reminiscent of the Rossby Wave Instability

we checked that the instability criteria was fulfilled and we confirmed the presence of vortices

the RWI at the edge of CBD

we performed a series of simulations for mass ratios between 1 and 0.1

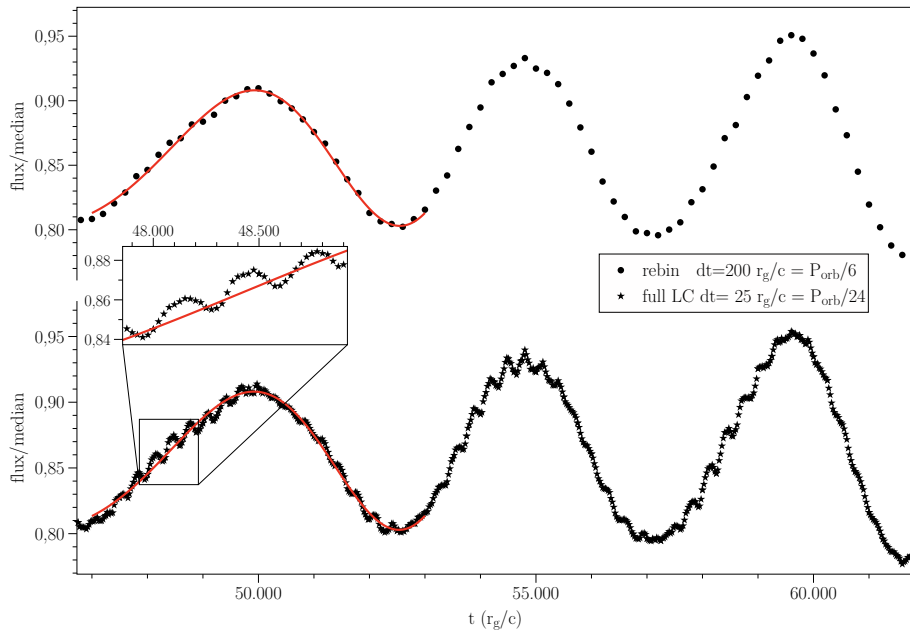
→ confirm that the RWI was indeed present



→ this is a first step into studying the RWI as a possible origin for the lump

→ if the association is confirmed it would make the RWI-induced variability a great observable for a large portion of pre-merger BBHs

from the RWI to the observable

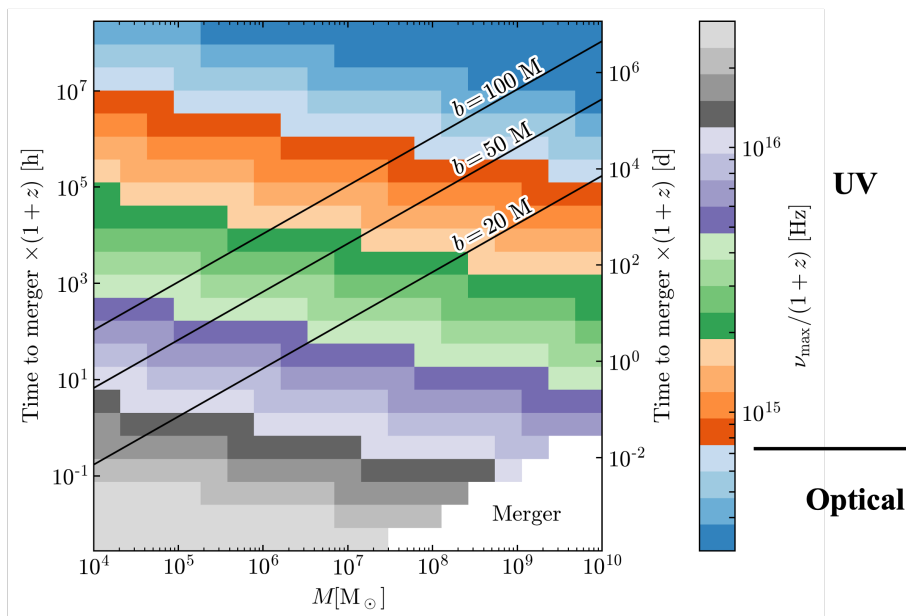


thanks to the criteria we know which system can trigger the RWI without needing a new simulation per system

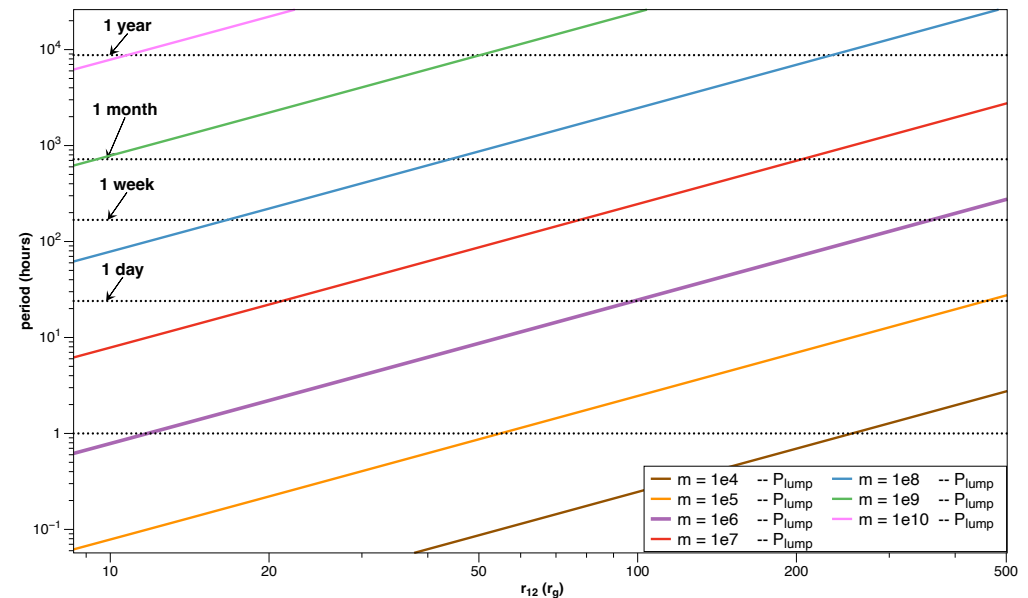
RWI-induced variability $\sim$ orbital velocity at twice the separation of the binary

with a good enough time resolution, we can also see the modulation related to the binary period

this modulation is most detectable in UV-Optical for SMBHs near the merger



with a period from one year to less than 1 hour depending on the binary mass and separation

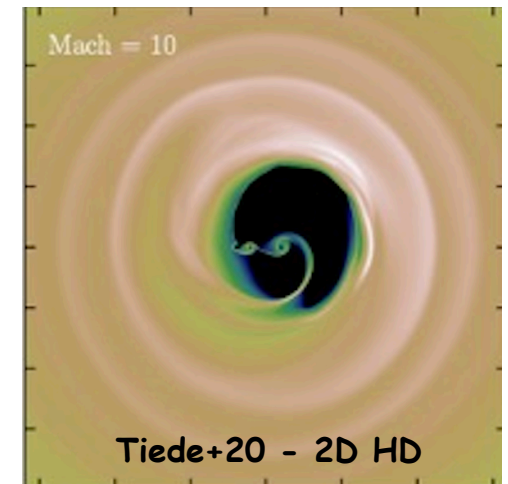
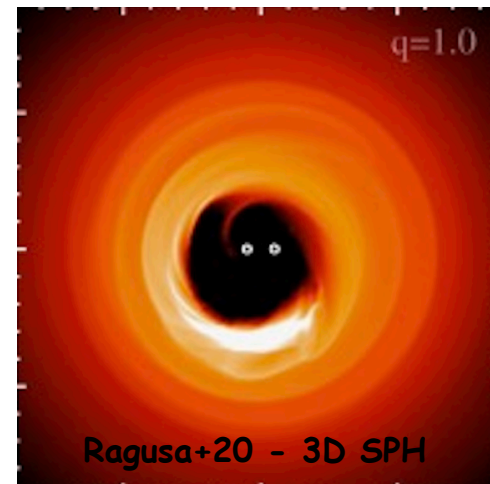
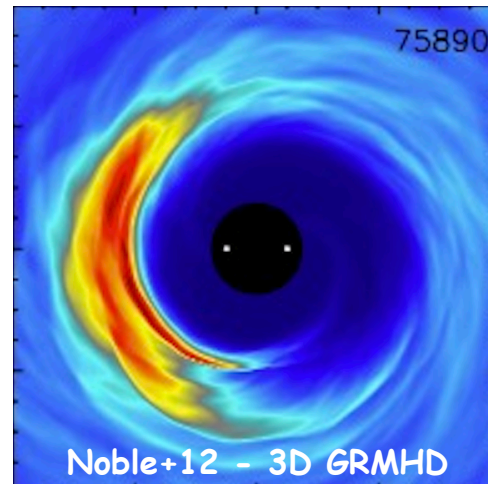
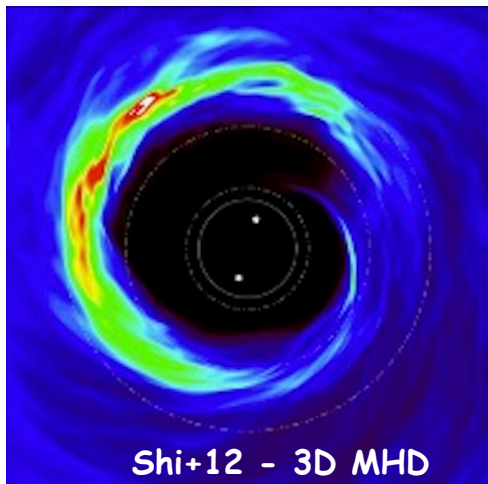


Finding BBHs thanks to the RWI at the edge of CBD

there is a need find the best observables to detect pre merger BBHs within the GW localization box

→ the circumbinary disk is the longest lived structure for near merger binary black holes

→ an orbiting lump is seen in numerical simulations near the edge of circumbinary disks



→ the edge of circumbinary disks are unstable to the RWI in near equal mass systems

→ making the predictable RWI-induced variability a great observable for a large portion of pre-merger BBHs

but that only works if the lump is indeed associated with the RWI and we validate its existence for a wider parameter space

the quest for lower mass ratios

in the binary black hole binary community it is commonly accepted that this lump does not exist if the mass ratio is less than 0.1 (sometimes included).

➡ but we see so many of those vortices in simulations of protoplanetary disks

which is interesting as there is a need to go to lower mass ratios to study EMRI and their link with SMBH quasi periodic flaring activity

→ but those simulations are expensive in terms of computing time

➡ this seems to be a good moment to bridge the study of the RWI over the mass ratio and physics from protoplanetary setup ($q \ll 1e-3$) to the binary black-hole setup ($1 < q < 0.1$) to show how robust the appearance of the RWI is.

➡ This conference seems to be a great place to start such collaborations.

if you have existing simulations with a low mass ratio (planet) with a horseshoe/vortex type of structure at the edge of the outer gap and want to contribute to that quest for lower mass ratios, please contact me!